COMP2611 COMPUTER ORGANIZATION
DATA REPRESENTATION

Numbers

- **Bits are able to represent both data and instruction in digital computers**

- **In this topic, you will learn:**
  - How to represent integer numbers (both signed and unsigned)?
  - How to represent fractions and real numbers?
  - What is a representable range of numbers in a computer?

- **To be covered in Computer Arithmetic topic:**
  - Arithmetic operations: How to add, subtract, multiply, divide integer numbers
  - How to build the hardware that takes care of arithmetic operations

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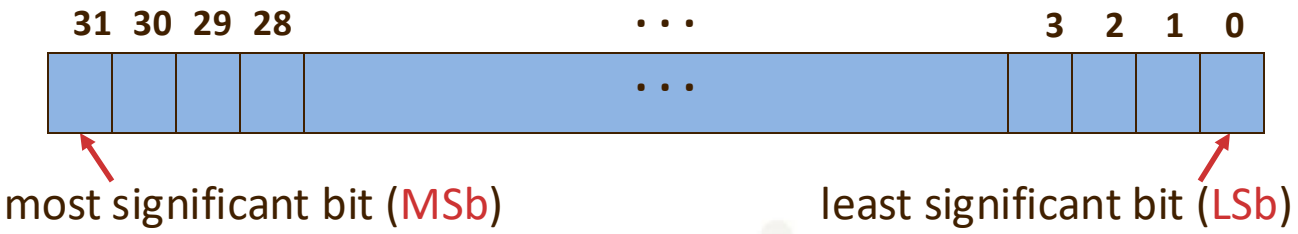
SIGNED AND UNSIGNED INTEGER

Review: Base, Representation and Value

- Numbers can be represented in any **base**
  - Decimal, binary, hexadecimal
- Positional notation helps to calculate the numerical value represented
- A sequence of bits (a.k.a. **bit sequence**) usually work together
- Bits are grouped and numbered 0, 1, 2, 3 … from right to the left:



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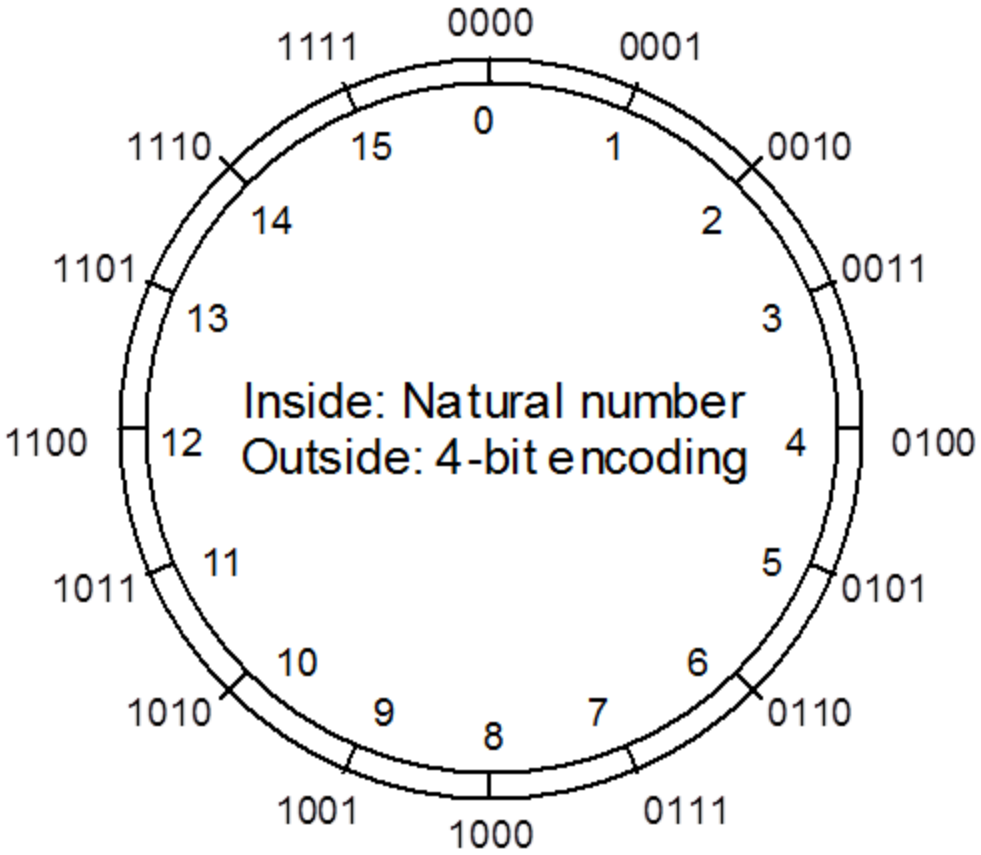
Unsigned Binary Integers

- Given k bits
- Min = $000.....00_2 = 0_{10}$
- Max = $111.....11_2 = 2^k - 1_{10}$
- Representable range is $[0, 2^k - 1]$



2's Complement

- **Most common scheme of representing negative numbers in computers**
- **Affords natural arithmetic (no special rules!)**
- **The most significant bit is called the sign bit**
  - 0 for non-negative, 1 for negative
- **The positive half uses the same representation as before**
- **To represent a negative number in 2's complement notation…**
  1. Decide upon the number of bits (n)
  2. Find the binary representation of the positive value in n-bits
  3. Flip all the bits (change 1's to 0's and vice versa)
  4. Add 1

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2's Complement Example

- What is the representation of -6 in 2's complement on 4 bits?

```
i)   Start from the representation of +6
```

$0110_2 = 6_{10}$

```
ii)  Invert bits to get 1's complement
```

$1001_2 = -7_{10}$

```
iii) Add 1 to get 2's complement
```

$1010_2 = -6_{10}$

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2's Complement Example (cont.)

- What is the representation of -6 in 2's complement on 8 bits?

i) Representation of +6	$0000\ 0110_2 = 6_{10}$
ii) Invert:	$1111\ 1001_2 = -7_{10}$
iii) Add 1	$1111\ 1010_2 = -6_{10}$

- What is the representation of -6 in 2's complement on 32 bits?

i) Start from the representation of +6

$0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0110_2 = 6_{10}$

ii) Invert bits to get 1's complement

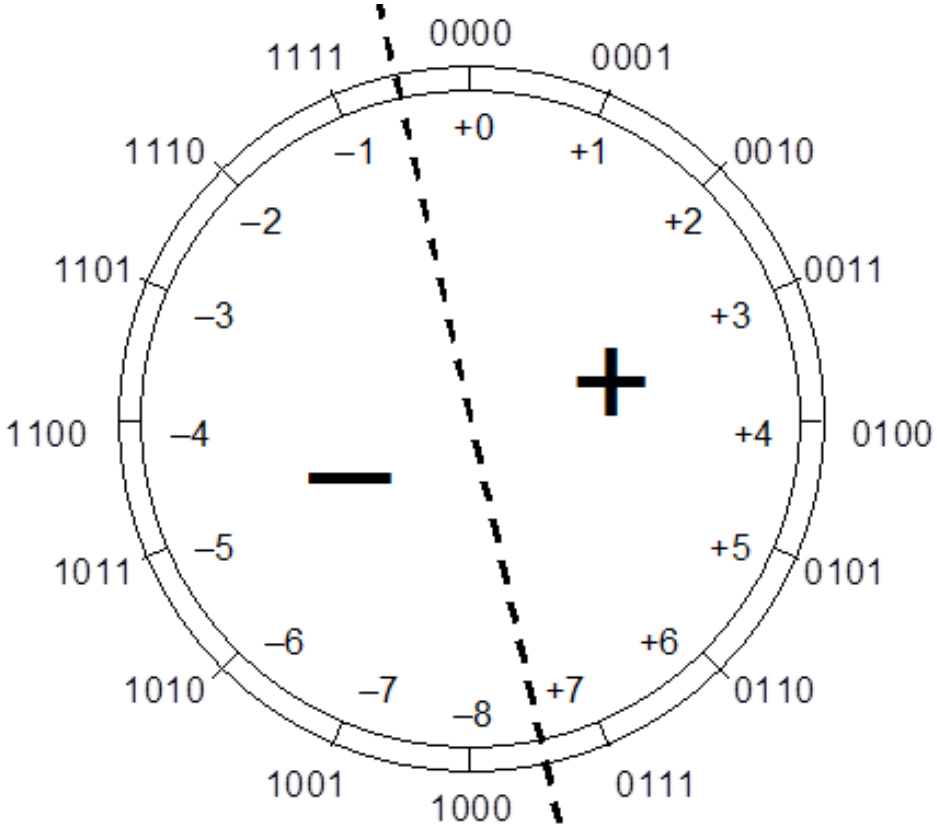
$1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1001_2 = -7_{10}$

iii) Add 1 to get 2's complement

$1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1010_2 = -6_{10}$

Signed Binary Integers

- **Given k bits**
- Min = $100.....00_2 = -2^{k-1}{}_{10}$
- Max = $011.....11_2 = 2^{k-1} - 1_{10}$
- **Representable range is** $[-2^{k-1}, 2^{k-1} - 1]$



Largest and Smallest Signed Integer

- Example: what is largest and smallest signed integer represented by 8 bits and 16 bits
- **largest**

$0111\ 1111_2 = 2^7 - 1 = 127$

$0111\ 1111\ 1111\ 1111_2 = 2^{15} - 1 = 32768 - 1 = 32767$

- **smallest**

$1000\ 0000_2$

Invert and add 1: $0111\ 1111_2 + 1 = 1000\ 0000_2 = 2^7 = 128$

=> - 128

$1000\ 0000\ 0000\ 0000_2$

Invert – add 1: $0111\ 1111\ 1111\ 1111_2 + 1 = 2^{15} = 32768$

=> - 32768

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Representable Range

- **All integers in range [smallest, largest] can be represented**
- **[smallest, largest] is the representable range**

- **Largest integer represented by a 32 bit word**
  - $0111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111_2 = (2^{31} - 1)_{10} = 2{,}147{,}483{,}647_{10}$
- **Smallest integer represented by a 32 bit word**
  - $1000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000_2 = -2^{31}{}_{10} = -2{,}147{,}483{,}648_{10}$

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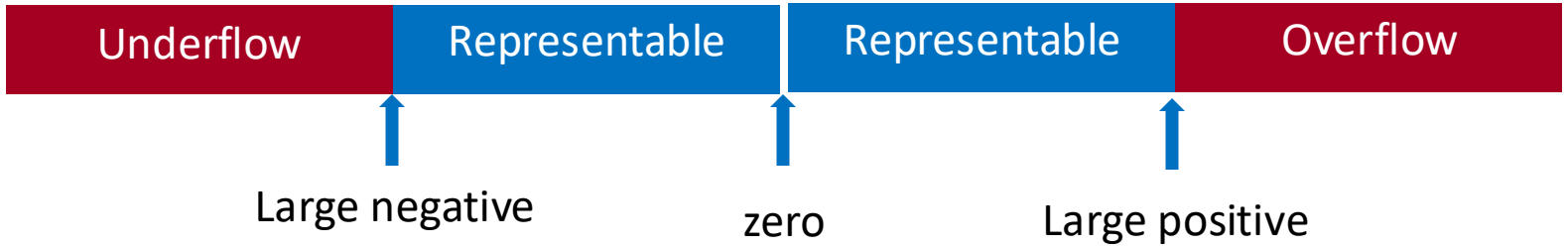
Overflow and Underflow in Signed Integer



- **Given the number of bits used in representing a signed integer**
  - **Overflow** (signed integer)
    - The value is bigger than the largest integer that can be represented
  - **Underflow** (signed integer)
    - The value is smaller than the smallest integer that can be represented

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Signed vs. Unsigned Integers

- **Signed**
  - **negative** or **non-negative** integers, e.g. `int` in C/C++
- **Unsigned**
  - **non-negative** integers, e.g. `unsigned int` in C/C++

- **Ranges for signed and unsigned numbers**
  - 32 bit words `signed`:
    - from

      $0111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111_2 = (2^{31} - 1)_{10} = 2{,}147{,}483{,}647_{10}$
    - to

      $1000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000_2 = -2^{31}{}_{10} = -2{,}147{,}483{,}648_{10}$
  - 32 bit words `unsigned`:
    - from

      $0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000_2 = 0_{10}$
    - to

      $1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111_2 = (2^{32} - 1)_{10} = 4{,}294{,}967{,}295_{10}$

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Sign Extension

- **Consider using a cast in C/C++ on a 32 bit machine**

```
int i;  /* signed integer represented on 32 bits */
char a; /* Character represented on 8 bits */
i = (int)a; /* What are the values of upper 24 bits in i? */
```

- **Similar things happen in hardware when an instruction loads a 16 bit number into a 32 bit register (hardware variable)**
  - Bits 0~15 of the register will contain the 16-bit value
  - What should be put in the remaining 16 bits (16~31) of the register?

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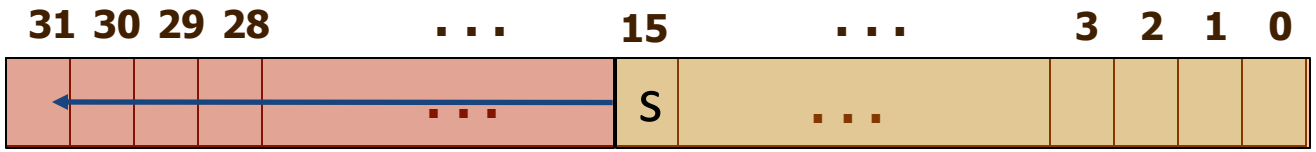
Sign Extension

- **Sign extension is a way to extend signed integer to more bits**
  - Check the sign bit, and extend it
  - If sign is 0 then fill with 0, If sign is 1 then fill with 1



- **Example: load a 16-bit signed value to a 32-bit register**
  - 2 (16 bits -> 32 bits):

0000 0000 0000 0010 -> 0000 0000 0000 0000 0000 0000 0000 0010

  - -2 (16 bits -> 32 bits):

1111 1111 1111 1110 -> 1111 1111 1111 1111 1111 1111 1111 1110

- **Does sign extension preserve the same value?**

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Zero Extension

- **Zero extension fills missing bits with 0**
- **Example:**
  - Bitwise logical operations (e.g. bitwise AND, bitwise OR)
  - Casting unsigned numbers to larger width

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FLOATING POINT NUMBERS

Why Floating Point?

- **Representation for non-integral numbers**
  - **Numbers with fractions** (called real numbers in mathematics)
    - e.g. 3.1416
  - **Very small numbers**
    - e.g., 0.00000000001
  - **Very large numbers**
    - e.g., 1.23456×10^{10} (a number a 32-bit integer can't represent)
- **In decimal representation, we have decimal point**
- **In binary representation, we call it binary point**

$$101.11_2 = (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (1 \times 2^{-2})_{10} = 5.75_{10}$$

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Scientific Notation & Normalized Scientific Notation

- **Scientific notation**
  - A single digit to the left of the decimal point
  - e.g. 1.23×10^{-3}, 0.5×10^{5}
- **Normalized scientific notation**
  - Scientific notation with no leading 0's
  - e.g. 1.23×10^{-3}, 5.0×10^{4}

- **Binary numbers can also be represented in scientific notation**
- **All normalized binary numbers always start with a 1**

$$1.xxx \ldots xxx_2 \times 2^{yyy \ldots yyy_2}$$

- **Example** $5.75 = 101.11_2 = 1.0111_2 \times 2^{10_2}$

- **Such numbers are called floating point in computer arithmetic**
- **Because the binary point is not fixed in the representation**

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Distribution of Floating Point Numbers

- $1.\underbrace{xxx\ldots xxx_2}_{\text{significand or mantissa}} \times 2^{\underbrace{yyy\ldots yyy_2}_{\text{exponent}}}$

- Example: 2-bit significand, exponent = {-2, -1, 0, 1}

e= -2	e = -1	e = 0	e = 1
1.00 X 2^(-2)=4/16	1.00 X 2^(-1) = 8/16	1.00 X 2^0 = 16/16	1.00 X 2^1 = 32/16
1.01 X 2^(-2)=5/16	1.01 X 2^(-1) = 10/16	1.01 X 2^0 = 20/16	1.01 X 2^1 = 40/16
1.10 X 2^(-2)=6/16	1.10 X 2^(-1) = 12/16	1.10 X 2^0 = 24/16	1.10 X 2^1= 48/16
1.11 X 2^(-2)=7/16	1.11 X 2^(-1) = 14/16	1.11 X 2^0 = 28/16	1.11 X 2^1 = 56/16

- Floating point representation is approximate arithmetic
- Finite range, limited precision

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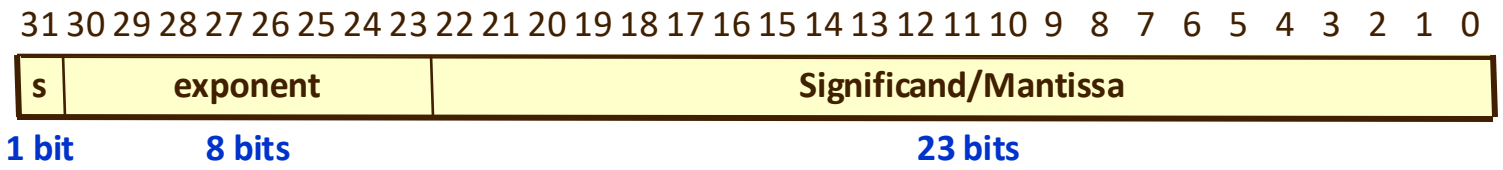
Single-Precision

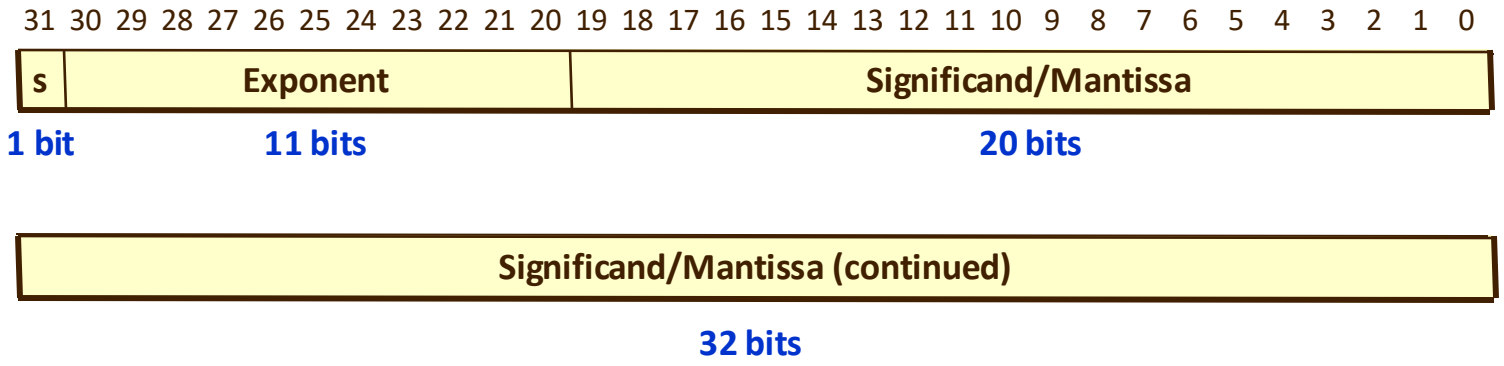
- Single-precision floating point uses **32-bit** sign-and-magnitude representation
- 8 bits for exponent, 23 bits for significand



- **Interpretation**
  - Roughly gives 7 decimal digits in precision
  - Exponent scale of about 10^{-38} to 10^{+38}

Double-Precision

- Double-precision floating-point uses 64 bits
- 11 bits for exponent, 52 bits for significand



- **Interpretation**
  - Provides precision of about 16 decimals
  - Exponent scale from 10^{-308} to 10^{+308}

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IEEE754 Floating Point Standard

- **Defined by IEEE in 1985**
- **Developed in response to divergence of representations**
  - Portability issues for scientific code
- **Now almost universally adopted**
- **Two representations**
  - Single precision (32-bit)
  - Double precision (64-bit)

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IEEE754 Floating-Point Format

	single: 8 bits double: 11 bits	single: 23 bits double: 52 bits
S	**Exponent**	**Significand**

$$x = (-1)^S \times (1 + Significand) \times 2^{Exponent-Bias}$$

- **Sign bit 0 ⇒ non-negative, 1 ⇒ negative**
- **Biased exponent**
  - Exponent field saves actual exponent + Bias, which ensures to be an unsigned value
  - Single: Bias = 127; Double: Bias = 1023
- **Significand with implicit 1**
  - Significand is normalized to be 1.xxxxx
  - There's always a leading 1 before binary point
  - The leading 1 is implicit (hidden bit) and not saved in significand field
- **Special cases are used to represent denormalized significand, NaN, etc.**

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Decimal -> IEEE754 Example

- **Give the IEEE754 representation of -0.75_{10} in single & double precisions**
- **Answer**
  - Scientific notation: $-0.75 = -0.11_2 \times 2^0$
  - Normalized scientific notation: $-1.1_2 \times 2^{-1}$
  - Sign = 1 (negative), exponent = -1

  - Single precision:

    S = 1, Significand = 100...00 (23 bits)

    Biased exponent = -1+127 = 01111110

    1 01111110 10000000000000000000000

  Double precision:

    S = 1, significand = 100...00 (52 bits)

    Biased exponent = -1+1023 = 01111111110

    1 01111111110 1000000000000000 000

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IEEE754 -> Decimal Example

- What decimal number is represented by this word (single precision)?

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- Answer:

$$
\begin{aligned}
&(-1)^s \times (1 + Significand) \times 2^{(Exponent - Bias)} \\
&= (-1)^1 \times (1 + 0.25) \times 2^{(129-127)} \\
&= -1 \times 1.25 \times 2^2 \\
&= -1.25 \times 4 \\
&= -5.0
\end{aligned}
$$

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Why Implicit 1 and Biased Exponent?

- **Implicit 1 packs even more bits into the significand**
  - The number is actually 24 bits long in single precision (implied 1 and 23-bit fraction), and 53 bits long in double precision (1 + 52)
- **Increase the precision of your numbers**
- **Biased exponent is for faster comparisons (for sorting, etc.), allow integer comparisons of floating point numbers**
  - Unbiased exponent
  - Biased exponent

1/2	0	1111 1111	000 0000 0000 0000 0000 0000
2	0	0000 0001	000 0000 0000 0000 0000 0000

1/2	0	0111 1110	000 0000 0000 0000 0000 0000
2	0	1000 0000	000 0000 0000 0000 0000 0000

Single-Precision Range (for Normalized Numbers)

- **Exponents 00000000 and 11111111 reserved**
- **Smallest value**
  - Exponent: 00000001
    $\Rightarrow$ actual exponent = 1 – 127 = –126
  - Fraction: 000…00 $\Rightarrow$ significand = 1.0
  - $\pm 1.0 \times 2^{-126} \approx \pm 1.2 \times 10^{-38}$
- **Largest value**
  - exponent: 11111110
    $\Rightarrow$ actual exponent = 254 – 127 = +127
  - Fraction: 111…11 $\Rightarrow$ significand ≈ 2.0
  - $\pm 2.0 \times 2^{+127} \approx \pm 3.4 \times 10^{+38}$

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Double-Precision Range (for Normalized Numbers)

- **Exponents 0000…00 and 1111…11 reserved**
- **Smallest value**
  - Exponent: 00000000001
    $\Rightarrow$ actual exponent = 1 – 1023 = –1022
  - Fraction: 000…00 $\Rightarrow$ significand = 1.0
  - $\pm 1.0 \times 2^{-1022} \approx \pm 2.2 \times 10^{-308}$
- **Largest value**
  - Exponent: 11111111110
    $\Rightarrow$ actual exponent = 2046 – 1023 = +1023
  - Fraction: 111…11 $\Rightarrow$ significand ≈ 2.0
  - $\pm 2.0 \times 2^{+1023} \approx \pm 1.8 \times 10^{+308}$

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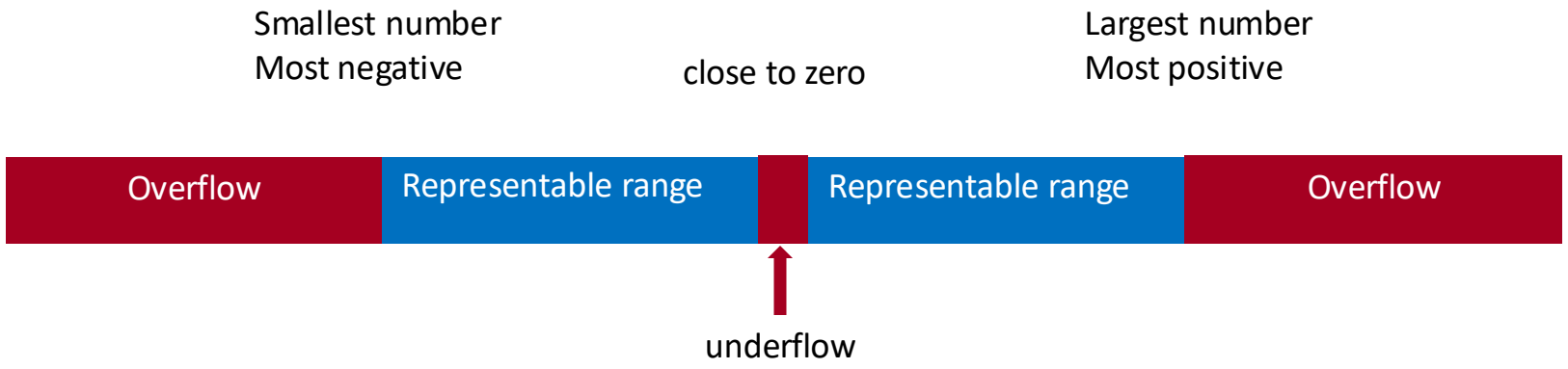
Overflow and Underflow in Floating Point



- **Overflow**
  - A positive exponent becomes too large to fit in the exponent field
- **Underflow**
  - A negative exponent becomes too large to fit in the exponent field

IEEE 754 Standard for Floating Point Arithmetic

Single precision:

Denormalized | Normalized | Special Cases

Significand \ Exponent	0	1 - 254	255
0	0	$(-1)^S \times (1.F) \times (2)^{E-127}$	$(-1)^S \times (\infty)$
≠ 0	$(-1)^S \times (0.F) \times (2)^{-126}$		non-numbers e.g. 0/0 , $\sqrt{-1}$

Double precision:

Significand \ Exponent	0	1 - 2046	2047
0	0	$(-1)^S \times (1.F) \times (2)^{E-1023}$	$(-1)^S \times (\infty)$
≠ 0	$(-1)^S \times (0.F) \times (2)^{-1022}$		non-numbers e.g. 0/0 , $\sqrt{-1}$

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Denormalized Numbers and Special Cases

Exponent	Significand	Usage
111…..11	000…..00	❑ ±Infinity ❑ Can be used in subsequent calculations, avoiding need for overflow check
111…..11	≠ 000…00	❑ Not-a-Number (NaN) ❑ Indicates illegal or undefined result ❑e.g., 0.0 / 0.0 ❑ Can be used in subsequent calculations
000…..00	000…..00	❑ $\pm$0.0
000…..00	≠ 000…00	❑ Hidden bit is 0 (no implicit 1) ❑ $x = (-1)^S \times (0 + Significand) \times 2^{-126}$ ❑ Even smaller than normal number

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IEEE754 Examples

0	00000000	00000000000000000000000 =	0
1	00000000	00000000000000000000000 =	-0
0	11111111	00000000000000000000000 =	+ infinity
1	11111111	00000000000000000000000 =	- infinity
0	11111111	01001100010001000001000 =	**NaN (Not a Number)**
1	11111111	01001100010001000001000 =	**NaN**
0	10000000	00000000000000000000000 =	$+(1.0_2)\times(2)^{128-127}=2$
0	10000001	10100000000000000000000 =	$+(1.101_2)\times(2)^{129-127}=6.5$
1	10000001	10100000000000000000000 =	$-(1.101_2)\times(2)^{129-127}=-6.5$
0	00000001	00000000000000000000000 =	$+(1.0_2)\times(2)^{1-127}=(2)^{-126}$
0	00000000	10000000000000000000000 =	$+(0.1_2)\times(2)^{-126}=(2)^{-127}$
0	00000000	00000000000000000000001 =	$+(2)^{-23}\times(2)^{-126}=(2)^{-149}$

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CHARACTERS

Communicating with People

- Characters are unsigned bytes e.g., in C++ Char
- Usually follow the ASCII standard
- Uses 8 bits unsigned to represent a character

b_7 →					0	0	0	0	1	1	1	1
b_6 →					0	0	1	1	0	0	1	1
b_5 →					0	1	0	1	0	1	0	1
Bits	b_4 ↓	b_3 ↓	b_2 ↓	b_1 ↓	Row ↓ / Column →	0	1	2	3	4	5	6	7
	0	0	0	0	0	NUL	DLE	SP	0	@	P	`	p
	0	0	0	1	1	SOH	DC1	!	1	A	Q	a	q
	0	0	1	0	2	STX	DC2	"	2	B	R	b	r
	0	0	1	1	3	ETX	DC3	#	3	C	S	c	s
	0	1	0	0	4	EOT	DC4	$	4	D	T	d	t
	0	1	0	1	5	ENQ	NAK	%	5	E	U	e	u
	0	1	1	0	6	ACK	SYN	&	6	F	V	f	v
	0	1	1	1	7	BEL	ETB	'	7	G	W	g	w
	1	0	0	0	8	BS	CAN	(	8	H	X	h	x
	1	0	0	1	9	HT	EM	)	9	I	Y	i	y
	1	0	1	0	10	LF	SUB	*	:	J	Z	j	z
	1	0	1	1	11	VT	ESC	+	;	K	[	k	{
	1	1	0	0	12	FF	FC	,	<	L	\	l	|
	1	1	0	1	13	CR	GS	-	=	M	]	m	}
	1	1	1	0	14	SO	RS	.	>	N	^	n	~
	1	1	1	1	15	SI	US	/	?	O	_	o	DEL

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Exercise

What does the following 32 bit pattern represent: 0x32363131

- **If it were a 2's complement integer**
- **If it were an unsigned number**
- **A sequence of ASCII encoded bytes: 2611**
  - Checking the ASCII table gives 0x32 = code for character '2', 0x36 = code for character '6', 0x31 = code for character '1', 0x31 = code for character '1'
- **A 32 bit IEEE 754 floating point number**
  - s= 0, E = 01100100, S = 01101100011000100110001
  - This is a normalized number so E is biased.

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Exercise

- **Consider building a floating point number system like the IEEE754 standard on 8 bit only, with 3 bits being reserved for the exponent.**
  - What is the value of the bias? **3**
  - What is the representation of 0? **0 000 0000**
  - What is the representation of -4?

    $-4 = -1.0 \times 2^2$

    S=1, F= 0 and the biased exponent must be

    E – 3 = 2 or E = +5

    So - 4 = 1 101 0000
  - What is the next negative value representable after – 4? **1 101 0001 = - 4.25**
  - What does the byte 1 111 1011 represent? **- NAN**
  - What is the representation of -∞? **1 111 0000**

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Concluding Remarks

- **2's complement representation for signed numbers**
  - Smallest and largest, representable range
- **Floating-point numbers**
  - Representation follows closely the **scientific notation**
  - Almost all computers, including MIPS, follow **IEEE 754 standard**
- **Single-precision floating-point representation takes 32 bits**
- **Double-precision floating-point representation takes 64 bits**
- **Overflow and underflow in signed integer and floating number**

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